RITIK KUMAR

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PROFILE

Machine learning researcher with hands-on deep learning experience in PyTorch, focused on building, benchmarking, and critically evaluating neural-network outputs on noisy, real-world observational data. Experienced in assessing prediction quality and robustness, identifying failure modes, edge cases, and outliers, and building large-scale data preprocessing and quality-selection pipelines. Has designed tasks and reviewed AI-generated outputs against explicit requirements, and regularly reads and implements ML research papers. Published researcher (peer-reviewed paper plus preprints); looking to apply this evaluation-focused approach to computer vision and visual AI systems.

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, GPyTorch, neural-network training, LLMs, Siamese Networks (shared-weight architectures), Physics-Informed Neural Networks (PINNs), Deep Gaussian Processes, ReFANN, MC Dropout, MCMC-based predictive uncertainty.
- **Evaluating Model Outputs:** Assessing prediction quality, accuracy and robustness; failure-mode and edge-case investigation; outlier analysis; uncertainty quantification; Gaussian NLL; model benchmarking and selection; sensitivity analysis, ablation studies, bootstrap analysis, 5-fold cross-validation.
- **AI Output Review:** Evaluating AI-generated code against written requirements, judging correctness and test adequacy, writing clear task specifications and criteria.
- **Data Preprocessing & Quality:** Large-scale filtering and selection pipelines (S/N and background cuts, catalogue cross-matching of ∼1M entries), automated extraction, parallel processing (23 workers) and HPC/SLURM.
- **Experiment Tracking & Tuning:** Optuna hyperparameter search, training-stability analysis, loss-function design, comparative evaluation of alternative modelling approaches.
- **Programming & Scientific Computing:** Python (NumPy, SciPy, Pandas, scikit-learn, Matplotlib), SQL, some C, Bash, Git/GitHub, Linux; probability and statistics, linear algebra, numerical methods.

EDUCATION

Indian Institute of Science (IISc), Bengaluru *Aug 2022 – Jul 2027 (expected)*
BS–MS (Research) in Physics CGPA: 7.5/10

Coursework: Machine Learning, Probability & Statistics, Linear Algebra, Matrix Theory, Stochastic Models, Computational Physics, Numerical Methods.

SELECTED MACHINE-LEARNING RESEARCH

ML Researcher – ReFANN, Siamese Networks, and Physics-Informed Learning *Jan 2026 – Present*
National Astronomical Observatory of Japan (NAOJ); Advisor: Prof. Maria G. Dainotti

- Investigated a PINN training failure in which derivative-based physics loss caused predictions to collapse towards a near-constant solution; tested alternative loss formulations, gradual loss-weight warm-up, EMA-based loss normalisation, and automatic restart strategies.
- Analysed the trade-off between physics-informed constraints and predictive robustness, comparing uncertainty reduction against outlier rates to identify cases where improvements in one metric did not translate into better overall reliability.
- Developed and benchmarked multiple deep-learning approaches for reconstructing missing regions in noisy, irregularly sampled GRB light curves, systematically evaluating prediction quality and robustness across hundreds of time series under a common framework.
- Implemented ReFANN with uncertainty-aware outputs and physics-inspired gradient regularisation, evaluating fitted-parameter uncertainty reduction and outlier behaviour across 522 usable time series.
- Developed a shared-weight Siamese network (a deep-learning architecture for comparing paired inputs) with reference–anchor fusion and MC Dropout, comparing uncertainty reduction and outlier rates to assess the reliability of reconstructed light curves.
- Optimised PINN hyperparameters using Optuna, with 80 trials per series and a 5-fold cross-validation objective; assessed model sensitivity through cross-validation and ablation experiments.
- Codes are available here.

ML Research Contributor – Deep Gaussian Processes *Apr 2025 – Dec 2025*
National Astronomical Observatory of Japan (NAOJ); Advisor: Prof. Maria G. Dainotti

- Contributed to a comparative study of seven reconstruction models on 521 GRB light curves, assessing uncertainty

reduction, reconstruction quality, and outlier behaviour.

- Implemented a two-layer variational **Deep Gaussian Process** in GPyTorch for reconstructing incomplete GRB X-ray light curves, using variational inference and an RBF kernel to model nonlinear temporal structure and quantify predictive uncertainty.

Research Project – Physics-Informed Neural Networks for PDEs *Jan 2025 – Apr 2025*

Indian Institute of Science (IISc), Bengaluru; Advisor: Prof. Rishita Das

- Implemented PINNs in PyTorch for 1D, 2D, and 3D Burgers' equations and 1D Euler equations, including the Sod shock-tube problem with sharp discontinuities.
- Compared numerical solutions with Runge–Kutta, Euler, and central-difference methods; used the CFL criterion to reason about stable time stepping.

AI OUTPUT EVALUATION EXPERIENCE

Freelance AI Coding Evaluator (Paid Training)

- Designed a testable bug-fix task for the open-source Click Python library, writing a clear problem statement and explicit evaluation criteria (behaviour, edge cases, regression tests) for Unicode-aware shell completion, case normalisation, and regression testing.
- Evaluated AI-generated code changes against task requirements, reviewing implementation correctness and whether the proposed tests adequately covered the specified behaviour.
- Refined test criteria to avoid unnecessarily restrictive assumptions while maintaining coverage of the required functionality.
- Built and tested a Docker environment for a sample Node.js application, including repository cloning, dependency installation, and reproducible execution.

DATA-INTENSIVE RESEARCH: OBSERVATIONAL ASTROPHYSICS

AGN Spectral Modelling – ECAP Collaboration *Jan 2026 – Sep 2026*

Erlangen Centre for Astroparticle Physics (FAU), Germany Advisor: Dr. Thomas Dauser & Prof Joern Wilms

- Perform statistical model fitting and quality assessment, MCMC inference, and model comparison on multi-epoch XMM-Newton and NuSTAR data for X-ray spectral analysis of active galactic nuclei; develop Python analysis pipelines and run computationally intensive workloads on HPC using parallel processing. **Manuscript in preparation.**

Tracing of Outflows in Active Galactic Nuclei – Bachelor's Thesis *Sep 2025 – Apr 2026*

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Built an automated Python data pipeline to cross-match approximately one million catalogue entries with XMM-Newton observations, applying filtering and automating spectral extraction.
- Used 23-worker parallel processing and developed custom XSTAR grids to study complex X-ray absorption features and multi-phase ultra-fast outflows. **Manuscript submitted to The Astrophysical Journal; under review.**

Spectral Analysis of Seyfert Galaxies *Dec 2024 – Jan 2026*

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Built a Python pipeline to cross-match 30,000 sources with data reduction and S/N selection; performed broadband NuSTAR and XMM spectral modelling in XSPEC to measure coronal temperatures. **Manuscript in preparation.**

PUBLICATIONS

- Ayush, **Ritik**, Dainotti et al. (2026). Physics-Informed Neural Networks and Data-Driven Models for GRB X-ray Light-Curve Gap Reconstruction. arXiv. *Preprint; under review.*
- Kaushal, A., Dainotti, M.G., **Ritik**, et al. (2026). Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves. *Journal of High Energy Astrophysics (JHEAp).* DOI
- **Ritik**, Stalin, C.S., Reeves, J.N. Possible detection of unambiguous outflows in a Quasar. Submitted to ApJ, 2026.
- *In preparation:* **Ritik**, Stalin, C.S., Hota, J. Variation in Coronal Temperature in Seyfert-type AGN, 2026.

CONFERENCES, WORKSHOPS & ACHIEVEMENTS

- XRISM International Conference 2025, Japan – Certificate. ASI Conference 2026, Guwahati – (Certificate, Poster).
- **Internship, NAOJ Tokyo (Summer 2025)**; collaboration with Prof. Maria G. Dainotti since April 2024 and with IIA Bengaluru since Dec. 2024; **full member, ECAP X-ray Research Group, Germany** since Jan. 2026; **Management Coordinator**, Pravega 2024–25, IISc.